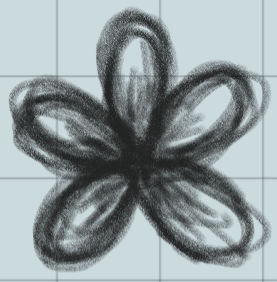
- Charles K. Alexander, Matthew N.O. Sadiku. Fundamentals of Electric Circuits.
- J. W. Nilsson, S. A. Riedel, Electric Circuits
- Prof. Doktor Cevdet Aca, Elektrik devreleri Analiz.



* Elektronların akış yönünün zıddı Elektrik Akımı'nın yönüdür.

ELEKTRON BOŞLUKLARI elektrik akımının aksıdır.

elektronlar ileri kayıyor, boşluk ise geriye kayıyor.

* Potansiyel Fark Gerilim, Voltaj denir

Akım → Amper

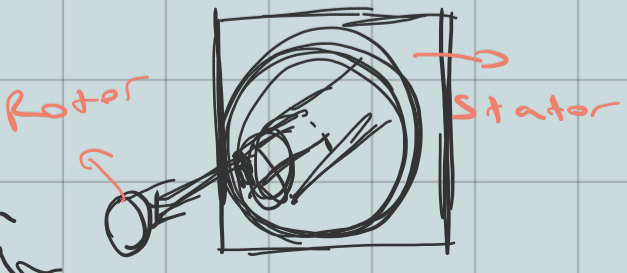
Gerilim → Volt

Elektrik Enerjisi = Zaman x Güç

* Alternatör = Generator = Jeneratör

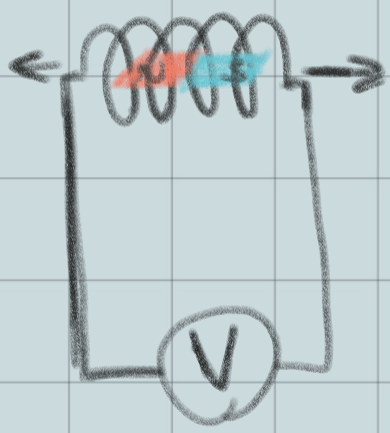
Elektrik Nasıl Üretiliyor?

1- Santrallerde elektriği Alternatör üretir. Rotor denen yapı döner. Rüzgar, su vb. bir güç ile döndürülür.



2- Stator sargılarında, Rotorun manyetik etkisi ile elektrik üretilir.

★ Küçük güçlü Alternatörlerde stator elektromıknatıs olarak çalışır ve elektrik gerilimi rotor sargılarında elde edilir.



Manyetik alan hareketli; iletken sabit.

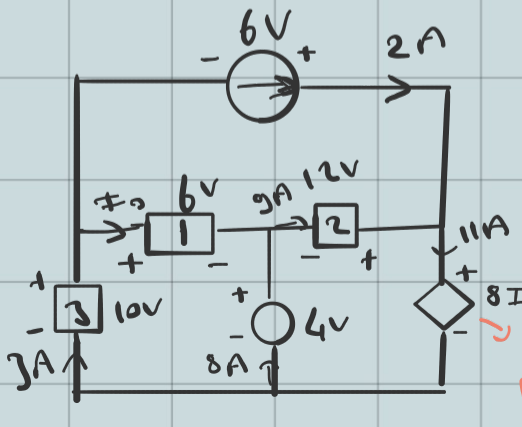


Manyetik alan sabit, iletken hareketli (Alternatörlerde bu şekilde çalışır)

$$\text{Güç} = \frac{\text{Enerji}}{\text{Saat}} \quad (\text{kwh})$$

★ Artı kısımdan akım giriyorsa güç tüketiyordur, Eksi kısmından giriyorsa güç veriyordur.

$I_o = ?$



$V_{yük} = V_{kaynak}$ gücü

$P = I \cdot V$

$P_1 = I_o \cdot 6\,W$ $\quad P_3 = 10 \cdot -3 = -30\,W$

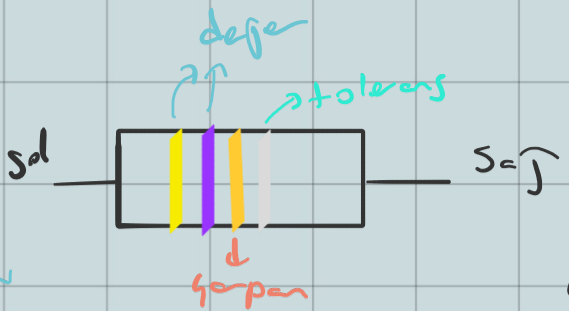
$P_2 = 12 \cdot -9 = -108\,W$ ⋮

$-12 + 6I_o - 108 - 30 - 72 + 176 = 0$

$I_o = 1A$

Direnç

$R = \rho \frac{l}{A}$ (özdirenç, tel uzunluğu, kesit alanı)



Toleransın renkleri:

Kırmızı: %2
Altın: %5
Gümüş: %10

(3) 47 000

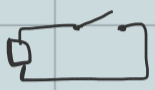
47 ± %10 kΩ

Siyah	0
Kahverengi	1
Kırmızı	2
Turuncu	3
Sarı	4
Yeşil	5
Mavi	6
Mor	7
Gri	8
Beyaz	9

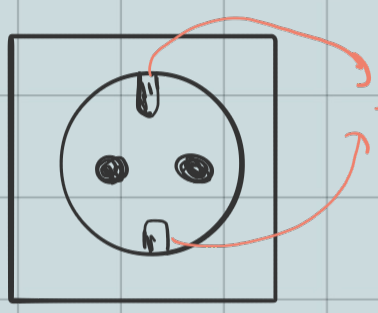
★ Kısa devrede direnç yok.



★ Açık devre sonsuz direnç sonsuz. (Açık devre)



Topraklama:



Multimetre: Ampermetre, Voltmetre, Ohmmetre vs. hepsini birden yapan cihaz.

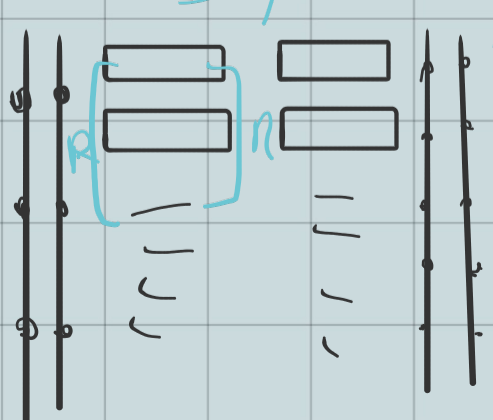


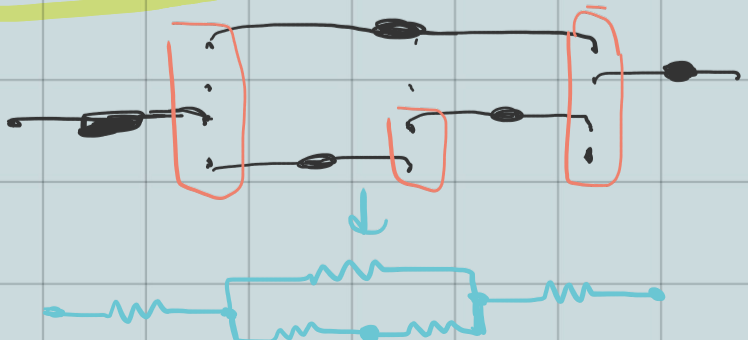
$\tilde{V}$ → Alternatif gerilim

$\bar{V}$ → Doğru gerilim

Breadboard

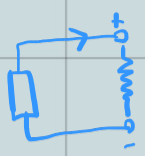
paralel bağlı dirençler





Ohm Kanunu

- $V = I \cdot R$
- $P = I \cdot V = (I)^2 \cdot R = \frac{V^2}{R}$



Artıdan giriyorsa güç tüketiyordur.

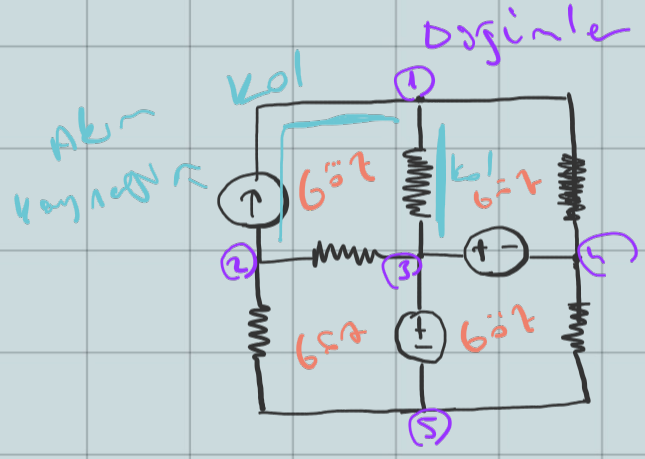
Kirchhoff Kanunları

Düğüm: İki ya da daha fazla devre elemanının birbirine bağlandığı noktalardır.

Çevre: Herhangi düğümden birden fazla geçilerek başlanan kapalı bir yoldur.

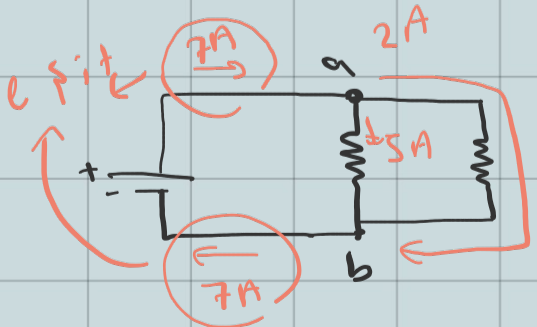
Kol: Yalnızca bir devre elemanı içeren ve elemanın her iki düğümünde olan devre parçasıdır.

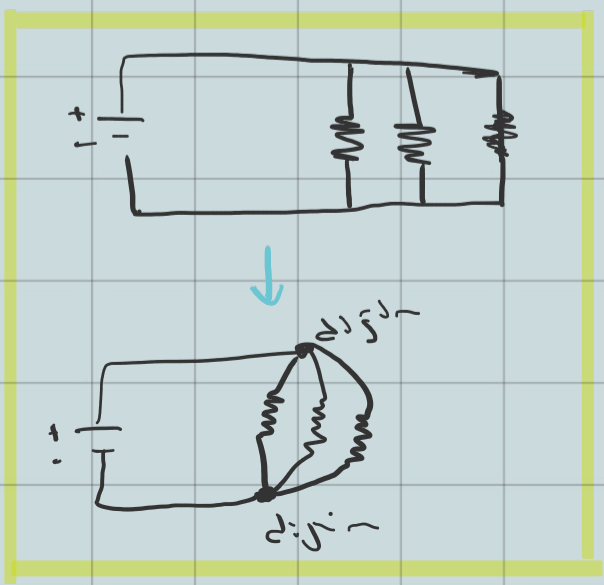
Göz: Herhangi bir kol tarafından bölünmeyen çevreye denir.

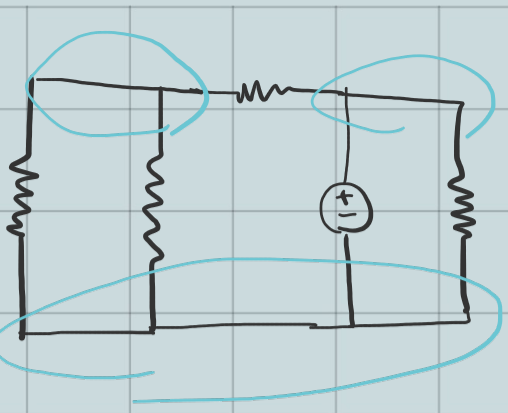


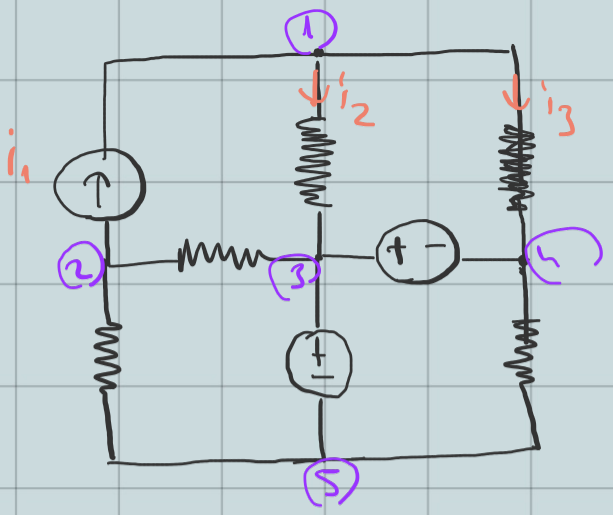
Kirchhoff'un Akımlar Kanunu (KAK)

Düğümden giren akımlarla çıkan akımlar eşit olmalı



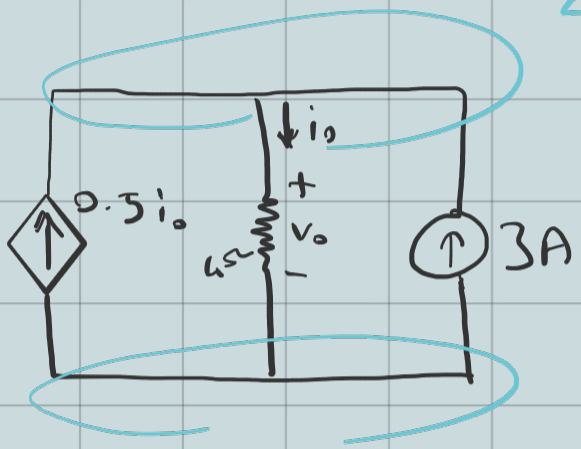






1. düğüm $-i_1+i_2+i_3=0$

giden + olan gelen -



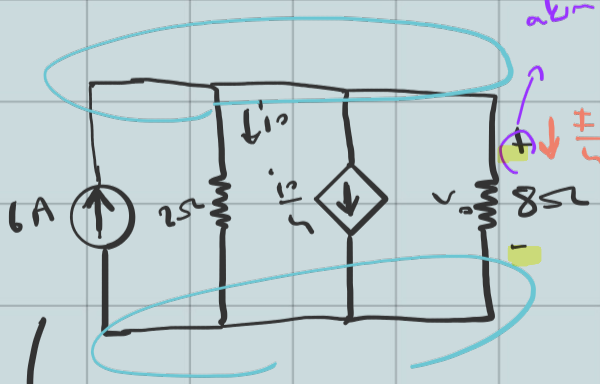
$3A+0.5i_o=i_o$

$\frac{3}{0.5}=i_o \quad i_o=6A$

$V_o=I\cdot R$

$V_o=6\cdot 4$

$=24V$



akım burdan geliyor demek

$V=I\cdot R$

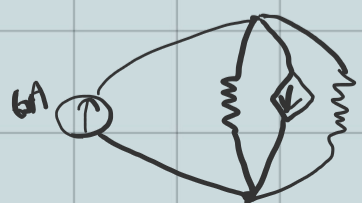
$2\Omega\cdot I_o=8\Omega\cdot a\cdot I_o$

$a=\frac{I_o}{4}$

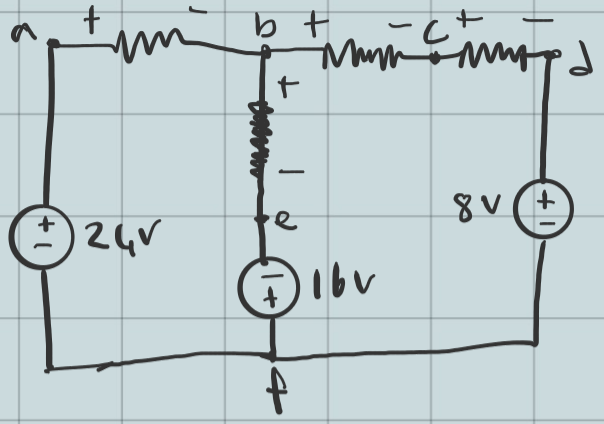
$6A=\frac{I_o}{4}+\frac{I_o}{4}+I_o$

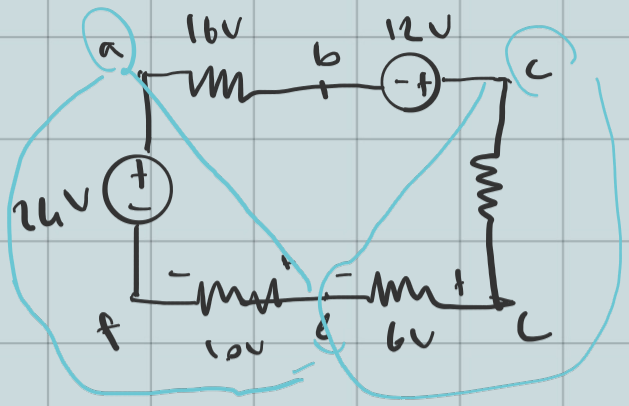
$6A=\frac{6I_o}{4}$

$I_o=4A$

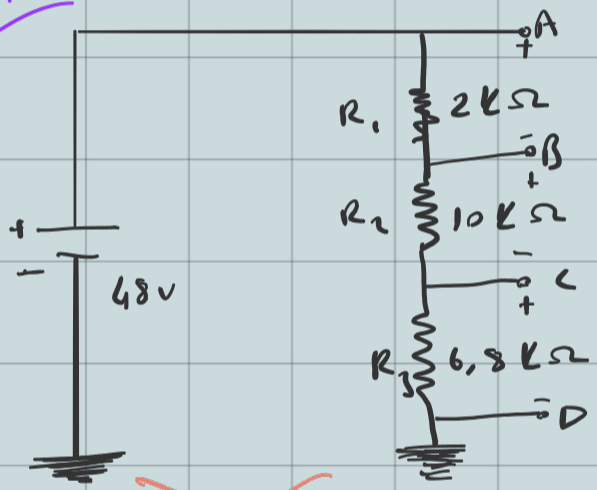


Kirchhoff'un Gerilimler Kanunu (KGK)





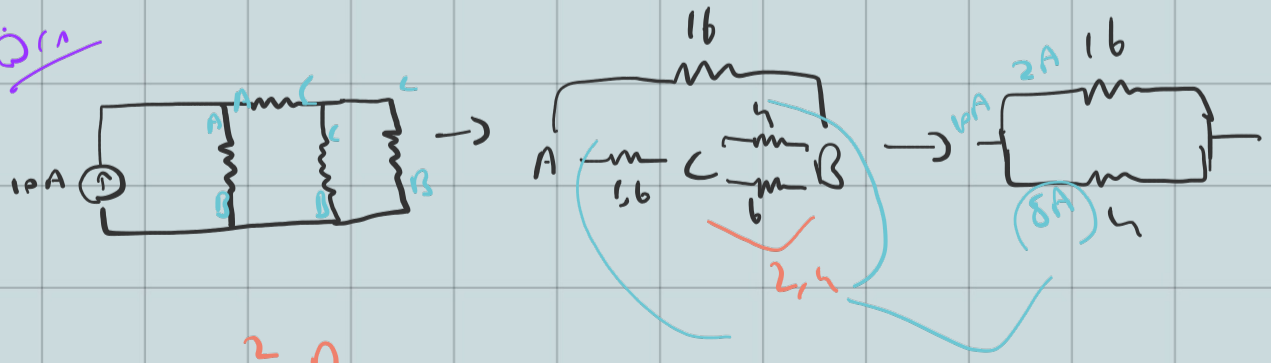
Örn



$R_T = 18,8$

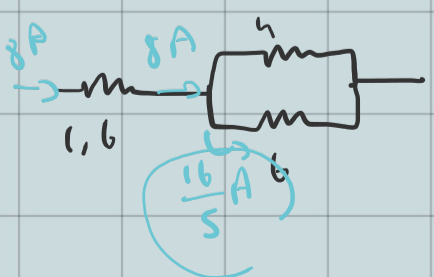
toprak(ana) (bağlı yerle aynı değerdedir)

Örn



$P = I^2 \cdot R$

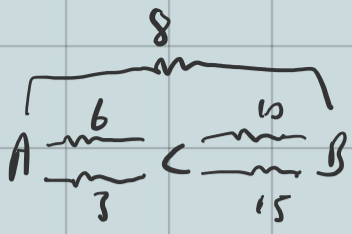
$= \frac{16 \cdot 16}{25} \cdot 6$



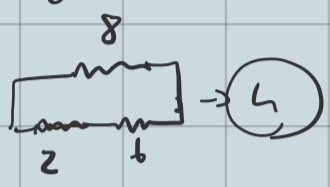
$8 \cdot \frac{4}{12} = \frac{16}{5} A$

Ser 1

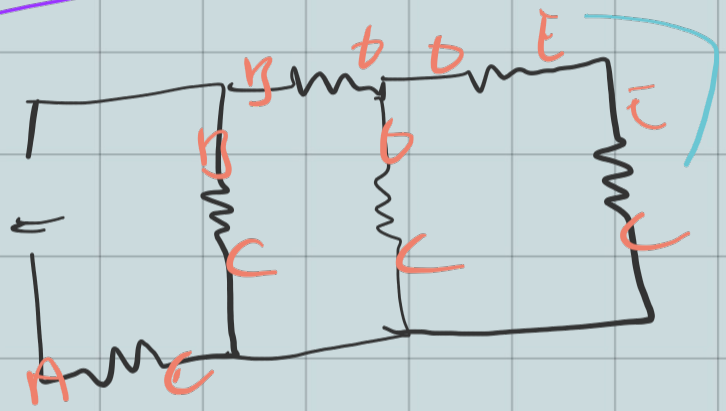




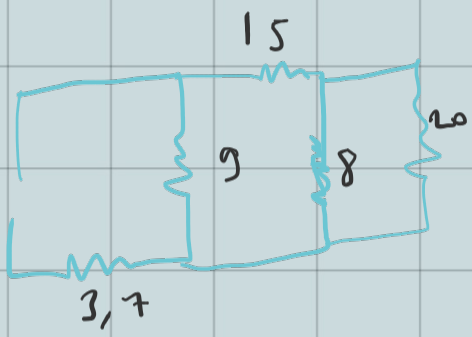
$\frac{15}{15} = (1)$



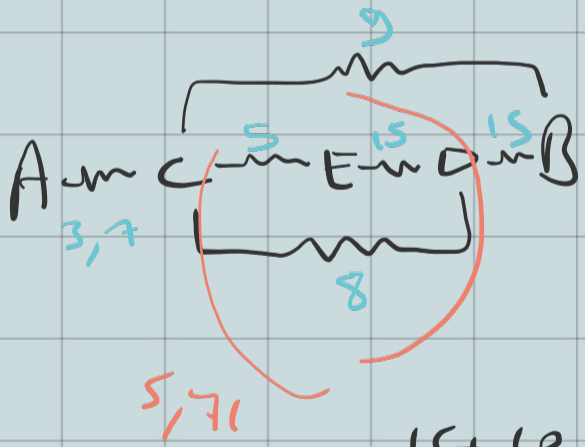
Ser 2



20Ω



$\frac{5}{\frac{20 \cdot 8}{28}} = \left(\frac{40}{7}\right)$ 5,7

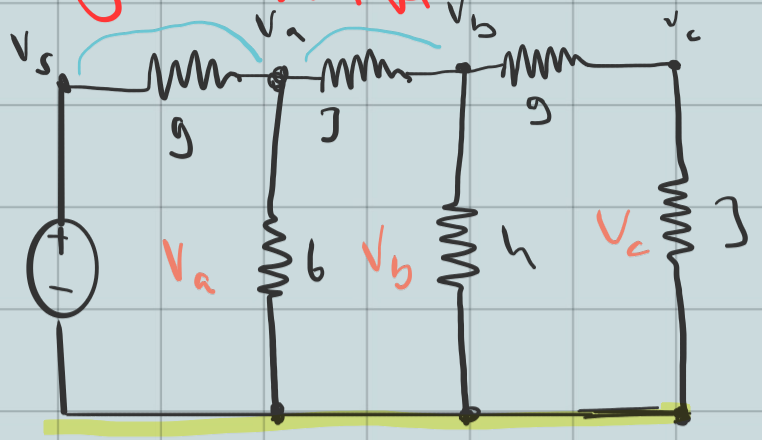


$\frac{\frac{40 \cdot 8}{7}}{\frac{40}{7} + 8} = \frac{10}{\frac{96}{7}} = \frac{10}{1} \cdot \frac{7}{96} = \frac{35}{12}$

$15 + \frac{10}{3} = \left(\frac{55}{3}\right) = \frac{10}{3}$

$\frac{\frac{55}{3} \cdot 3}{\frac{55}{3} + 3} = \frac{165}{82} = 165 \cdot \frac{3}{82}$

Düğüm Analizi

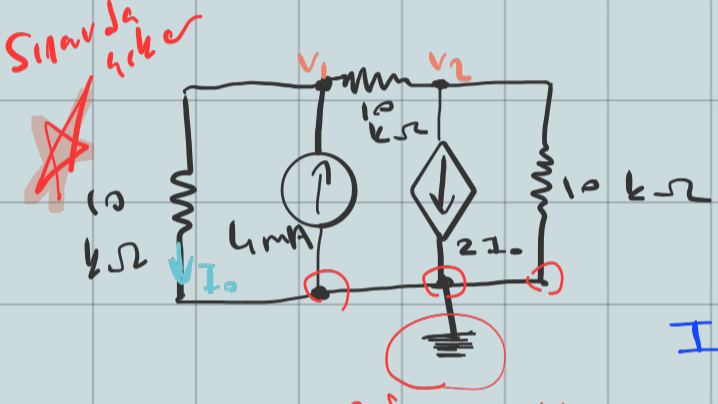


$$\frac{V_a - V_s}{9} + \frac{V_a}{6} + \frac{V_a - V_b}{3} = 0$$

$$\frac{V_b - V_a}{3} + \frac{V_b}{4} + \frac{V_b - V_c}{9} = 0$$

$$\frac{V_c - V_b}{9} + \frac{V_c}{3} = 0$$

Sınavda çıkar

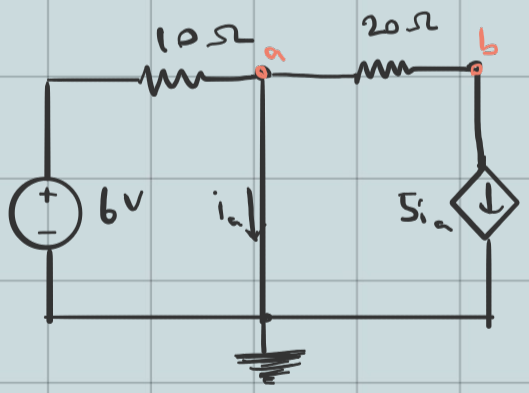


Düğüm V_1: $\frac{V_1}{10k} - 4mA + \frac{V_1 - V_2}{10k} = 0$

Düğüm V_2: $\frac{V_2 - V_1}{10k} + 2I_0 + \frac{V_2}{10k} = 0$

$I_0 = \frac{V_1}{10k}$

* referans noktasının üstü ve altı değişken olarak alınmaz.

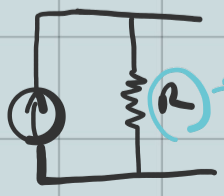


$\frac{V_a - 6}{10} + \frac{V_a - V_b}{20} + i_a = 0$

$\frac{V_b - V_a}{20} + 5i_a = 0$

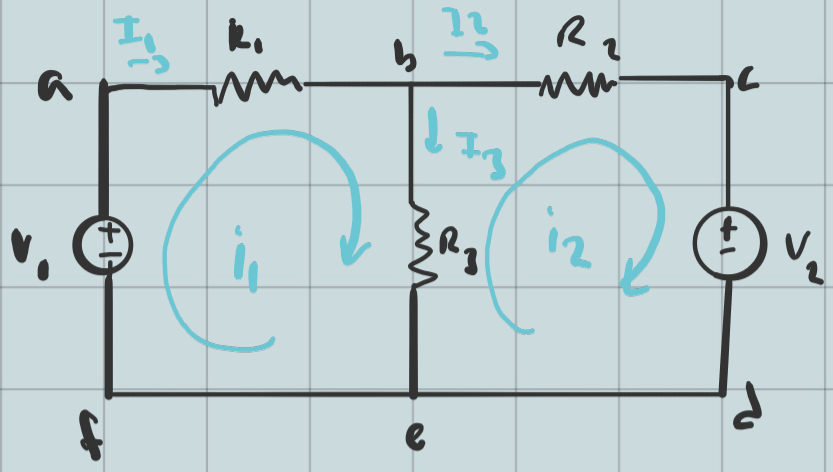
$V_a = 0V$

Akım Kaynakları

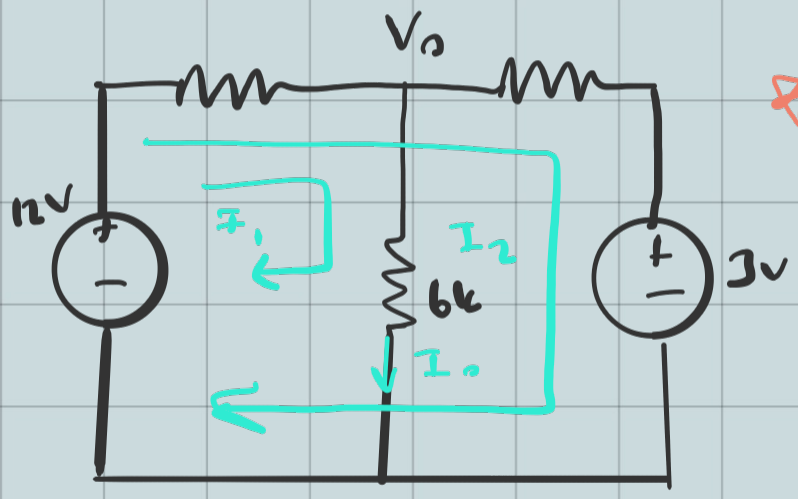


→ akım kaynağının iç direnci (paralel bağlıdır)

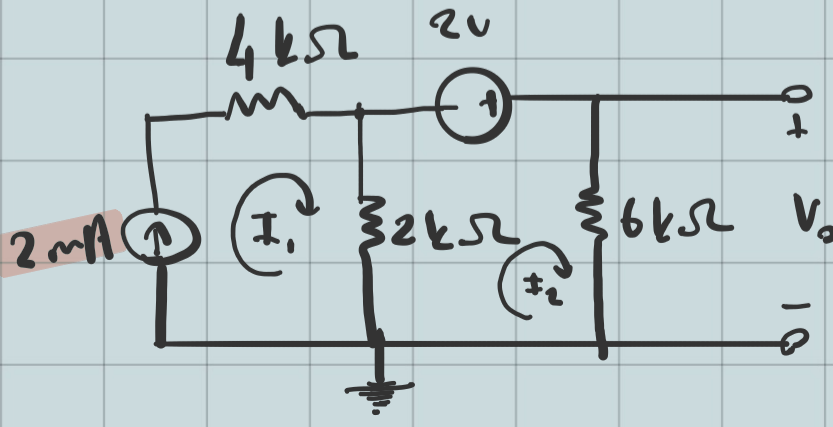
Çevre Analizi



Çevreler → abefa, bcdeb

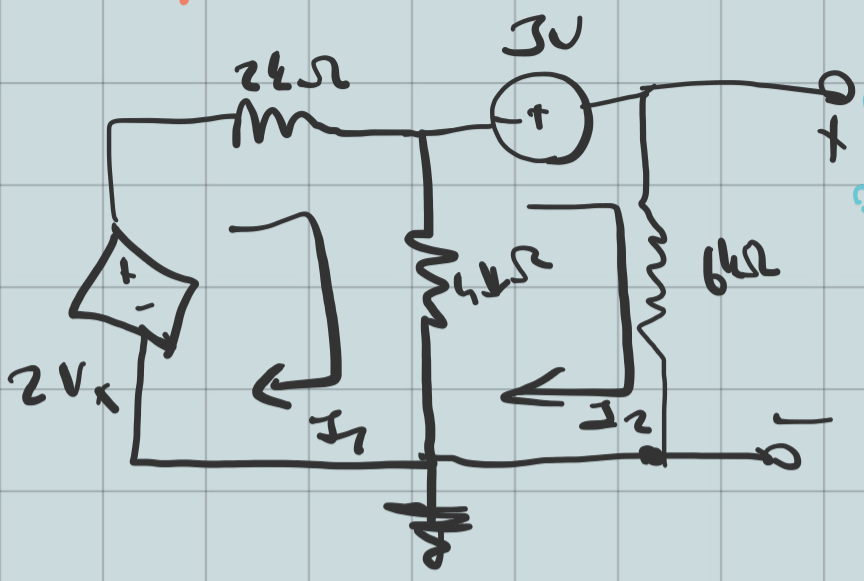


★ Gözleri kesen veya dış çevre seçmeyin



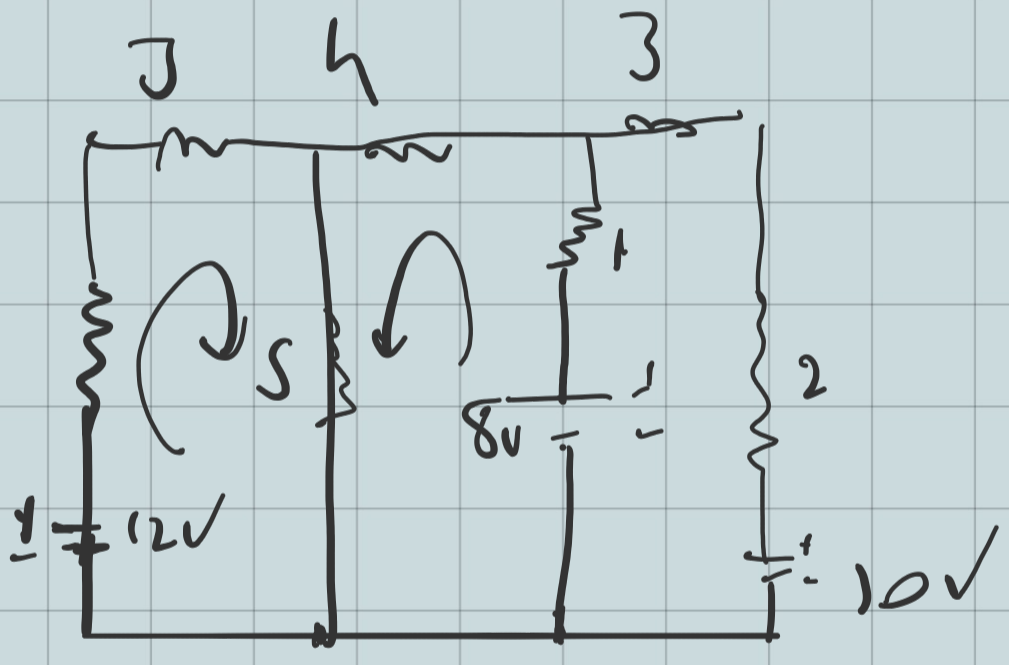
$I_1 = 2mA$

Süper Düğüm



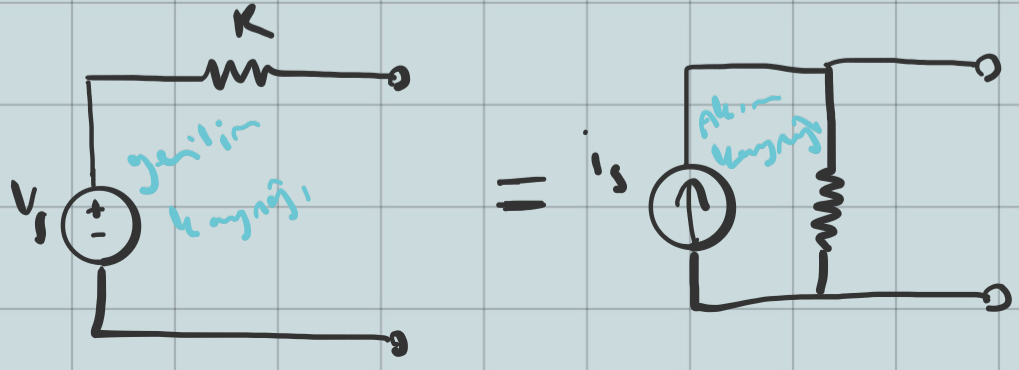
$0 = -2V_x + 2k \cdot I_1 + 4k(I_1 - I_2)$

$0 = -3V_x + 6k \cdot I_1 + 4k(I_2 - I_1)$



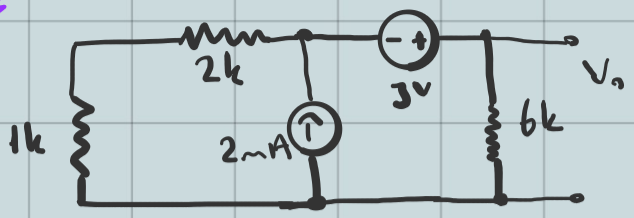
$0 = -12 + 10I_1 - 5I_2$

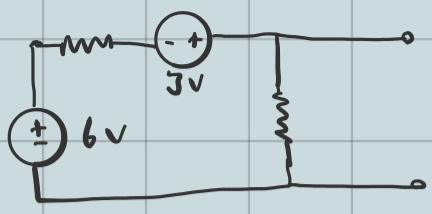
Süperpozisyon

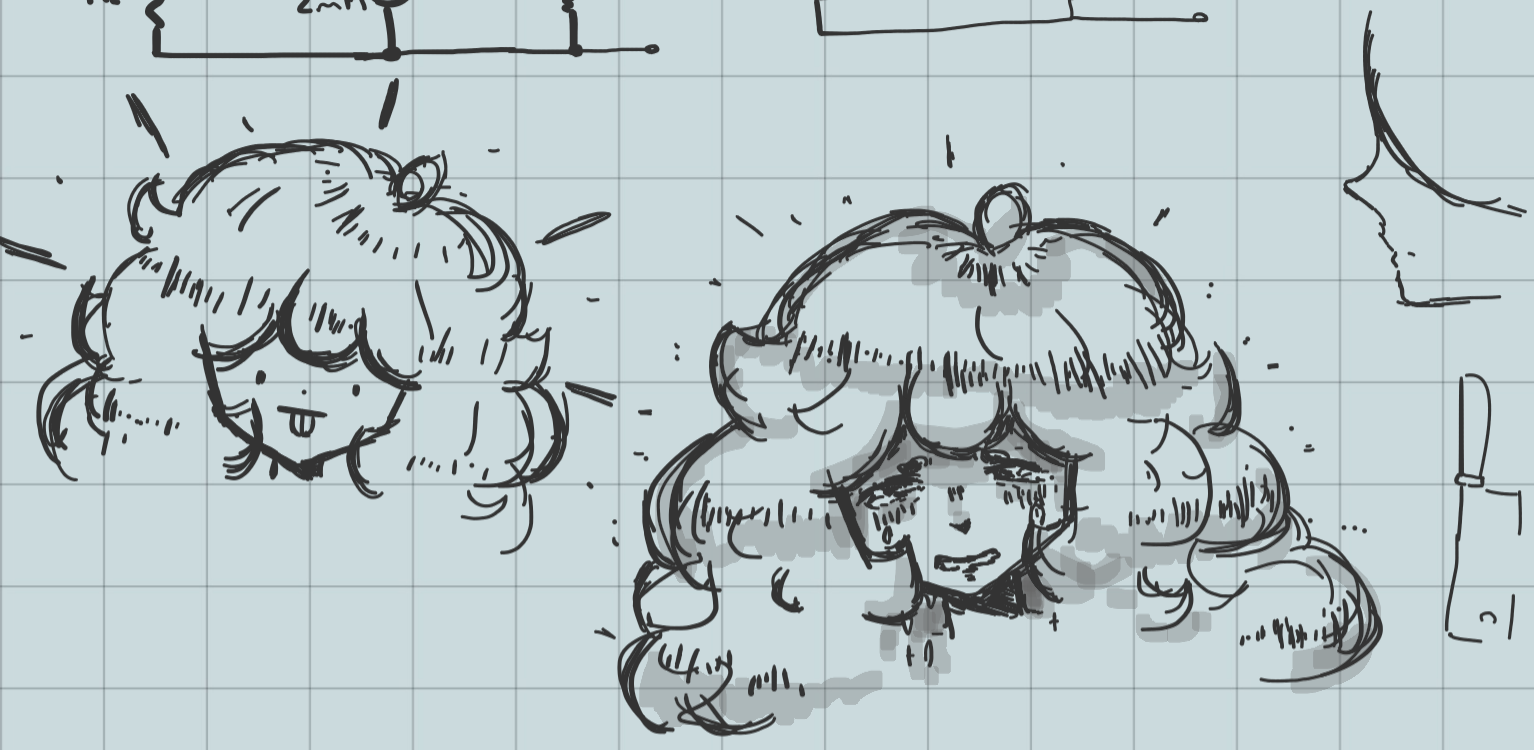


Ova

V_o sadrži blank

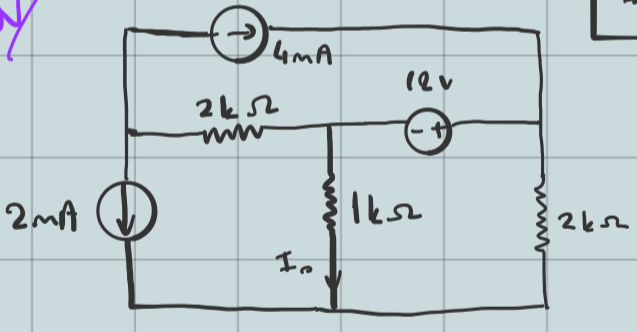






Ovaj

$I_o = ?$



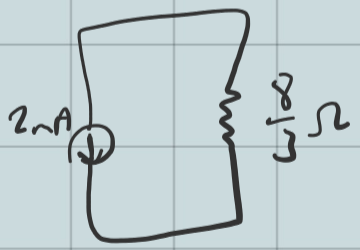
$$I_o' = 2 \cdot \frac{2}{3} = -\frac{4}{3}$$

2. yol

$$I_1 = I \cdot \frac{R_2}{R_1 + R_2}$$

$$R_{es_1} = \frac{1 \cdot 2}{3} = \frac{2}{3}$$

$$R_{es_2} = 2 + \frac{2}{3} = \frac{8}{3}\,\Omega$$



$$V = I \cdot R$$

$$= 2 \cdot \frac{8}{3} = \boxed{\frac{16}{3}}$$

$$V_{2k} = 2mA \cdot 2k = 4V$$

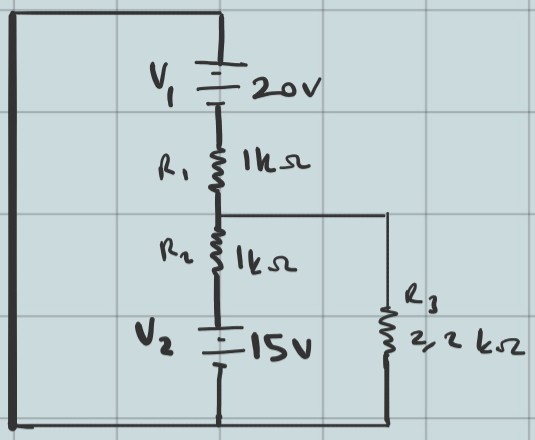
$$V_6 = \frac{16}{3} - 4 = \frac{4}{3}V$$

$$I_o' = V_6 \cdot 1k = \frac{4}{3}V \cdot 1k$$

$$= \boxed{-\frac{4}{3}mA}$$

Gergeyge

Örn

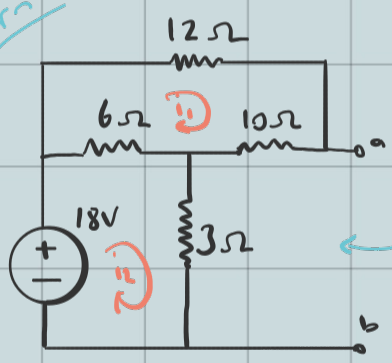




Thevenin ve Norton Teoremleri

Thevenin Yaklaşımı

Örn





$12i_1 + 10i_1 - 6(i_2 - i_1) = 0$

$6(i_2 - i_1) + 3i_2 - 18 = 0$

$i_1 = \frac{1}{2}$ A

$i_2 = \frac{7}{3}$ A

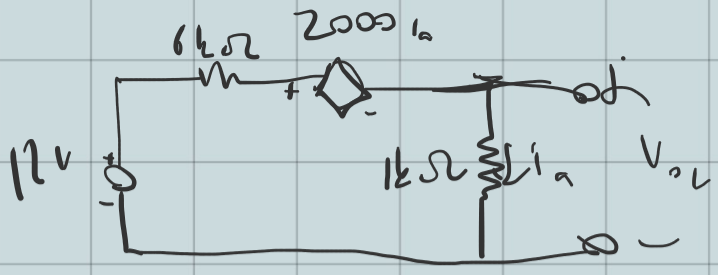
$V_{oc} = 3i_2 + 10i_1 = 3 \cdot \frac{7}{3} + 10 \cdot \frac{1}{2} = 12$ V

$R_1 = \frac{12(10+2)}{12+(10+2)} = 6\,\Omega$

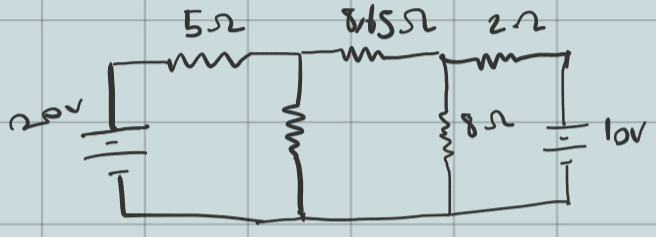
Yalnızca Bağımsız Kaynak içeren devrelr

Devreye test gerilimi uygulanır.

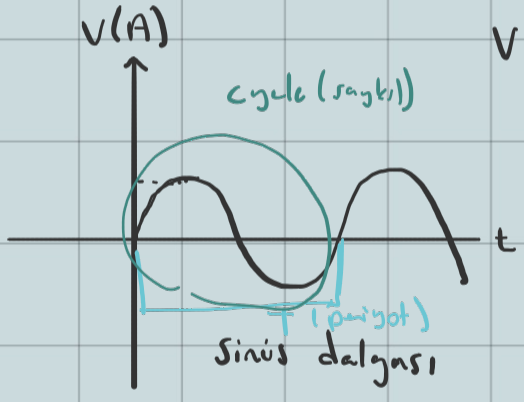




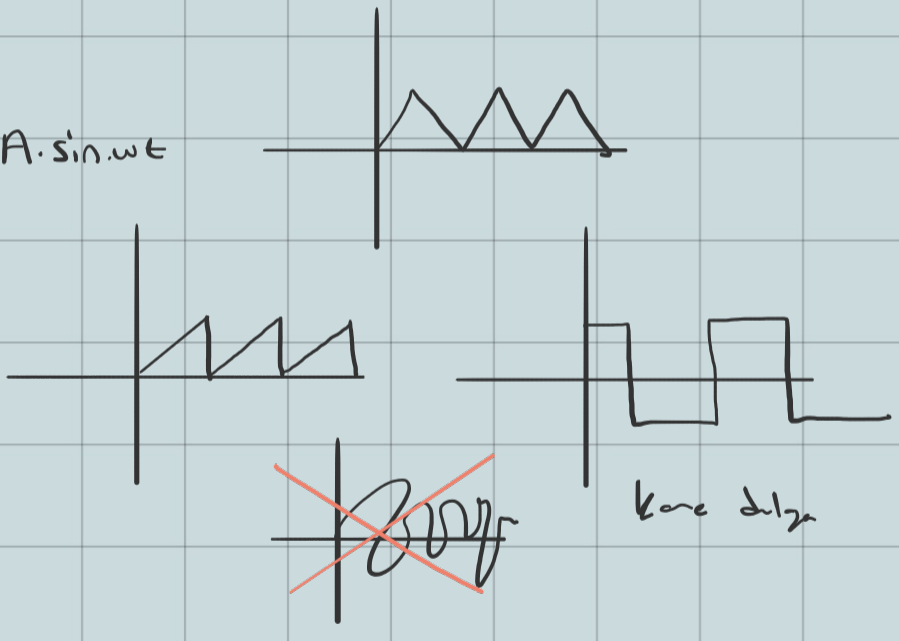
$-12 + 6i_a + 2000i_a + 1i_a$

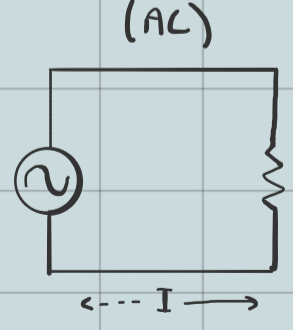


Alternatif Akım



$V(t) = A \cdot \sin \omega t$





Açısal Hız ω :3

Ani Değer Sinüs ve türdeki sürekli değişen Alternatif akım/gerilimin herhangi bir t anındaki değeri

$i = I_{max} \cdot \sin \omega t$

$v = V_{max} \cdot \sin \omega t$

$$V_a = \frac{1}{T} \int_0^T v(t)\,dt \quad \text{(birimi: Volt [V])}$$

Etkin Değer

220 V Alternatif akımın Etkin değerdir.

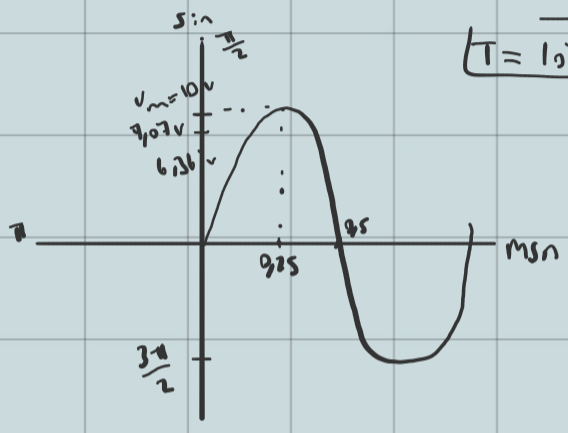
$V_{eff} = 0{,}707 \cdot V_m$

$I_{eff} = 0{,}707 \cdot I_m$

Tanımı: $V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$ Birimi: volt[V]

Tepe değeri V_p olan sinüs biçimli gerilim için: $V_{rms(sin)} = \frac{1}{\sqrt{2}} V_p$

Örn

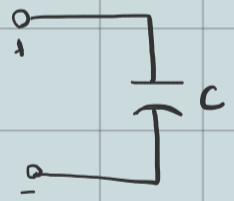


$T = 10^{-3}$ sn

Sinyalin denklemi

$u = U_m \cdot \sin\omega t = 10 \cdot \sin(2\pi \cdot$

Kapasitörler



$C = \varepsilon \frac{A}{d}$

C: farad(F)

$Q = C \cdot V$

Sabit kapasitans için

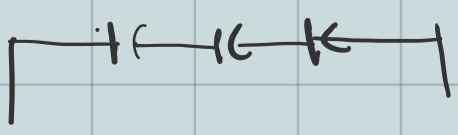
$i = C\frac{dv}{dt}$

$v = \frac{1}{C}\int_{-\infty}^{t}$

$w(t) = \frac{1}{2} C v^2(t)$

Seri bağlıysa, dirençlerde paralelken eşdeğer direnç bulduğumuz gibi bulduğumuz.

$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$



paralelken de dirençlerdeki gibi



$C_T = C_1 + C_2 + C_3$

Kapasitif Reaktans



Jade



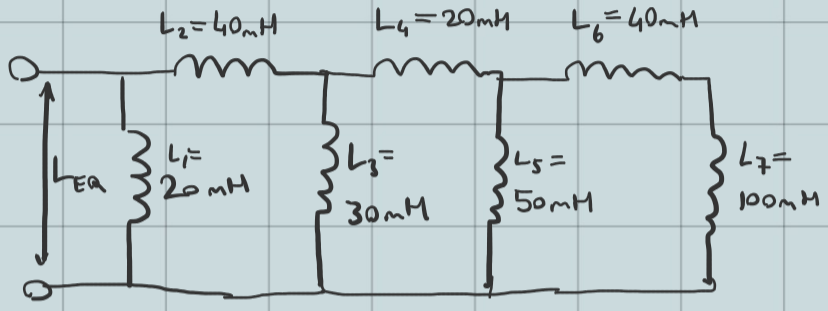
Isabelle



Endüktörler

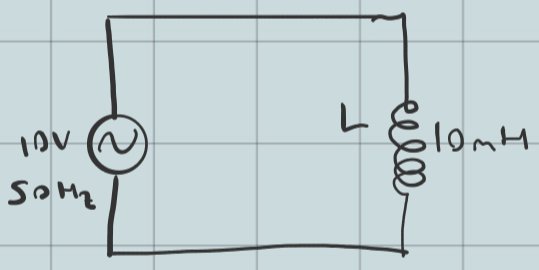
$$V(t) = L \frac{di(t)}{dt}$$

Edgar endüktör?



$$L = \frac{L_5 \times (L_6 + L_7)}{L_5 + L_6 + L_7} = \frac{50 \times (40 + 100)}{50 + 40 + 100} =$$

Endüktif Reaktans



- $X_L = 2\pi fL =$
- $X_L =$



Kapasitor

$$i(t) = C \frac{dv(t)}{dt}$$

Endüktör

$$v(t) = L \frac{di(t)}{dt}$$

$$\frac{1}{2\pi fC}$$

"90° ...